A sharp mechanical cost for a finite-duration reversal

A return with prescribed endpoint velocities $+u$ and $-u$ and acceleration ceiling a requires duration at least $2u/a$. Its minimum kinetic action is $mu^3/(3a)$, independent of any additional waiting time. Meanwhile its sampled polygon action converges with a sharp quadratic mesh bound. The endpoint reversal cost and the refinement error therefore have different limits.

1. Model and sharp minimum (C043)

Fix $m, a, T > 0$ and $0 < u < c$ in a specified inertial frame. The admissible paths are $X \in W^{2,\infty}(0,T)$, with continuous velocity representative $v = \dot{X}$,

$$X(0) = X(T) = 0, \quad v(0) = u, \quad v(T) = -u, \quad |v| \leq u, \quad |\dot{v}| \leq a \quad \text{almost everywhere.}$$

The functional is the kinetic cost $S_K[X] = (m/2)\int_0^T v^2 dt$, in action units. The acceleration is an admissible external control, with force $m\dot{v}$. This is a bounded-control mechanics problem; a prescribed autonomous potential, its potential contribution to Hamilton's action and a dynamical momentum receiver would specify additional physical data.

Proposition. The class is nonempty exactly when $T \geq 2u/a$. For such T, the unique minimizing velocity is, with $\tau = u/a$,

$$v_*(t) = \begin{cases} u - at, & 0 \leq t \leq \tau, \\ 0, & \tau \leq t \leq T - \tau, \\ -a(t - T + \tau), & T - \tau \leq t \leq T. \end{cases} \qquad S_{K,\min} = \frac{mu^3}{3a}.$$

Proof. Lipschitz continuity gives $2u = |v(T) - v(0)| \leq aT$. For a feasible duration the intervals $[0,\tau]$ and $[T-\tau, T]$ have disjoint interiors. On the first, $v(t) \geq u - at \geq 0$; on the last, $v(t) \leq -u + a(T-t) \leq 0$. Hence

$$\int_0^T v^2 dt \geq \int_0^\tau (u - at)^2 dt + \int_{T-\tau}^T [u - a(T-t)]^2 dt = \frac{2u^3}{3a}.$$

The displayed v_* has zero total integral and therefore returns to the starting position. It obeys both ceilings and attains equality. Equality forces each endpoint ramp and zero velocity in the middle almost everywhere; continuity gives uniqueness. Integration with $X(0) = 0$ fixes the path. The maximum excursion is $u^2/(2a)$, reached at the waiting segment.

At $T = 2u/a$ the velocity is $u - at$ throughout. Indeed equality in the total velocity-change bound forces $\dot{v} = -a$ almost everywhere, so the whole admissible class is then a singleton. At longer durations the admissible class includes variations even though the minimizing path remains unique.

2. Which premises hold the cost above zero?

The bound depends on prescribed nonzero endpoint speed and a finite acceleration ceiling. At fixed force ceiling $F_{\max} = ma$ it becomes

$$S_{K,\min} = \frac{m^2u^3}{3F_{\max}}.$$

For fixed m, a, T, choosing successively smaller positive u eventually preserves feasibility and gives $S_{K,\min} \to 0$. An upper speed ceiling allows this family. Likewise, increasing the permitted acceleration at fixed m, u, T drives the minimum to zero. Uniform positivity over a preparation class requires corresponding lower bounds on mass and endpoint speed and an upper bound on acceleration, or another premise controlling their combination.

For $T > 2u/a$, choose a nonzero smooth function ϕ supported strictly inside the waiting interval, with $\int \phi = 0$. Such functions are obtained by differentiating a nonconstant smooth compactly supported bump. For small $b \neq 0$, $v_b = v_* + b\phi$ satisfies the same speed/acceleration bounds and all endpoint conditions. Since the supports of ϕ and the nonzero part of v_* are disjoint,

$$S_K[v_b] - S_K[v_*] = \frac{mb^2}{2} \int \phi^2 dt \longrightarrow 0.$$

Thus the positive minimum total cost coexists with positive excess costs accumulating at zero. Its source is the endpoint-conditioned motion, rather than a discrete spacing of admissible action values. The cost is frame-specific: the endpoint velocities and return condition already select that frame.

3. The sampled polygon error (C044)

For any partition $\pi : 0 = t_0 < \cdots < t_N = T$, write $d_i = t_{i+1} - t_i$, $|\pi| = \max_i d_i$, and define the chord kinetic action

$$S_\pi = \frac{m}{2} \sum_i \frac{[X(t_{i+1}) - X(t_i)]^2}{d_i}.$$

For every admissible path, and more generally every velocity with Lipschitz constant at most a,

$$0 \leq S_K - S_\pi \leq \frac{ma^2}{24} \sum_i d_i^3 \leq \frac{ma^2 T}{24} |\pi|^2.$$

Proof. The chord velocity on each cell is its average $\bar{v}_i$. Square completion and the pair-variance identity give

$$S_K - S_\pi = \frac{m}{2} \sum_i \int_{t_i}^{t_{i+1}} (v - \bar{v}_i)^2 dt,$$

$$\int_I (v - \bar{v}_I)^2 dt = \frac{1}{2d} \int_I \int_I [v(s) - v(t)]^2 ds\, dt \leq \frac{a^2}{2d} \int_0^d \int_0^d (s-t)^2 ds\, dt = \frac{a^2 d^3}{12}.$$

Summing proves the claim. The constant $1/24$ is sharp: the feasible minimum-duration return has affine velocity with slope $-a$ on every cell, and attains the first upper bound for every partition. Equal-width cells also attain the mesh bound. In the longer-duration minimizer, cells contained inside a ramp saturate the local bound and cells inside the waiting interval have zero error.

At fixed m, a, T this convergence is uniform over the admissible class. For families with increasing acceleration bound a_N, the estimate still forces the defect to vanish if $a_N|\pi_N| \to 0$ (fixed m, T). Maintaining a positive defect along shrinking meshes therefore requires loss of this uniform acceleration control. This is a necessary condition, not a construction of a positive remainder.

4. Source comparison and next mechanical step

Liberzon's double-integrator example provides the bounded-acceleration control framework and a minimum-time problem with running cost one. Here the endpoint data and running cost $mv^2/2$ are different; the endpoint-envelope proof above establishes the exact result. B21 records the bounded prior-art search.

The next mechanical question is to realize finite-duration turns with a specified conservative receiver or interaction, then derive its correlation law. A07's external-control class provides a benchmark for such a realization. In the companion G02 task, the receiver's internal degrees of freedom also test which modes the available observables detect.

Run `python3 scripts/bounded_acceleration_checks.py` for exact integrals, matching conditions, variations and rational partition tests. The written proofs establish the general minima and uniform convergence.